AA–C–296
MARCH 5, 1956

SUPERSEDING
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FEDERAL SPECIFICATION

CHAIR, STUDENT, METAL AND WOOD

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers the component materials and their assembly into an adult-type, student, tablet arm chair for general classroom use.

1.2 Classification.

1.2.1 *Grade, type, and classes.* — The chair shall be of one grade and type, and of the following classes as specified in the invitation for bids (see figs. 1 and 2 and 3.2.3.1).

Class 1.—Chair with small table arm.
Class 2.—Chair with large table arm.

1.2.2 *Color of finish (metal).*—The color of finish of metal parts shall be gray and, unless otherwise specified, shall be the manufacturers' commercial standard color (see 3.4.1).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

P–P–556—Polish; Metal.
T–T–911—Twine; Jute.
NN–B–591 — Boxes; Fiberboard, Wood-Cleated (for Domestic Shipment).
NN–B–621—Boxes; Wood, Nailed and Lock-Corner.
TT–C–595 — Colors; (for) Ready Mixed Paints.
TT–E–489—Enamel; gloss, synthetic (for Exterior and Interior Surfaces).
UU–C–843 — Cushioning Material, Cellulosic.
LLL–B–631—Boxes; Fiber Corrugated, (for Domestic Shipment).
LLL–B–636—Boxes, Fiber, Solid (for Domestic Shipment).
MMM–A–188—Adhesive; Urea-Resin Type (Liquid and Powder).
PPP–B–585—Boxes; Wood, Wirebound.
PPP–B–601 — Boxes; Wood-Cleated-Plywood.

Military Specifications:

JAN–P–103—Packaging and Packing for Overseas Shipment — Boxes, Wood-Cleated, Solid Fiberboard.
JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.
MIL–B–107—Boxes, Wood, Wirebound (Overseas Type).
JAN–P–108—Boxes, Fiberboard, Corrugated and Solid (Military Type).
JAN–P–138—Packaging and Packing for Overseas Shipment—Boxes, Wood, Fiberboard-Lined.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking for Shipment and Storage.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

Department of Commerce Publication:

Commercial Standard CS–35—Hardwood Plywood.

(Copies of the Commercial Standard CS–35 may be obtained upon application to the Commodity Standards Division, Office of Industry and Commerce, U. S. Department of Commerce, Washington 25, D. C., or upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price may be obtained from the Superintendent of Documents.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Metal parts.*—All metal parts shall be of furniture steel having smooth surfaces which are free from excessive rust, pits, scale, burrs, sharp edges and corners.

3.1.1.1 *Glides.*—Glides shall be constructed of live-rubber cushion securely contained in die-formed $1\frac{1}{16}$ inch O.D. steel base of 0.050 inch minimum thickness with an 8-32 machine screw, hexagonal nut and lock washer. Screw shall have a $\frac{3}{8}$ inch minimum projection from top of rubber cushion. Steel base shall be casehardened to a minimum depth of 0.005 inch and test not less than 15N85 Rockwell. Base shall be nickel or chrome plated, other metal parts shall be zinc or cadmium plated. There shall be no metal-to-metal contact between metal base and equipment upon application.

3.1.1.2 *Rivets, bolts, screws, lockwashers and nuts.*—All rivets shall be slotless oval head type. All bolts, rivets, screws, lockwashers, and nuts shall be cadmium or zinc plated.

3.1.2 *Plywood.*—All plywood shall be of a commercial grade CS–35 hardwood plywood of balanced construction having exposed face plies of grade 1, birch, beech, cherry or maple, not less than $\frac{1}{28}$ inch thick. Plywood shall be bonded with urea resin or equivalent adhesive of sufficient strength and moisture resistance to pass the shear tests as specified in 4.3.1. Edges of all plies shall be smooth and sound, having no objectionable holes or crevices. No notches, holes or attaching hardware shall be visible in the top surface or edges of the work surface (tablet arm). Face veneer shall be matched for grain and color. Interior and bottom face veneer shall be not less than grade 2.

3.1.3 *Alternate arm.*—When specified, the work surface (tablet arm) may, alternatively, be made of built-up strips of clear and sound good commercial grade beech, birch, cherry or maple, joined with urea resin or equivalent adhesive of sufficient strength to pass the block shear tests as specified in 4.3.2.

3.2 Construction.

3.2.1 *Design and performance.*—The chairs shall be rigidly constructed of a die-formed sheet steel frame with wood arms, backs and seats to assure compliance with the test requirements of 4.3, and be essentially of the modern design and of the dimensions shown in figures 1 and 2.

3.2.1.1 *Tolerances.*—The following tolerances shall apply:

Dimensions shown on figure 2—plus or minus $\frac{1}{2}$ inch unless otherwise indicated.

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Dimensions, including thickness, of any plywood or hardwood parts—plus or minus 1/16 inch.

3.2.2 *Component parts.*—Each chair covered by this specification shall consist of the following components:

a. Metal parts:
1. Two side frames
2. A storage compartment or a front rail and a storage compartment
3. Front and rear arm supports
4. Four glides
5. Rivets, screws, bolts, nuts and lock washers and back rail as required

b. Wood parts
1. An arm
2. Back supports
3. A seat

3.2.3 *Metal parts.*

3.2.3.1 *Side frames.*—Each side frame shall be deep drawn stamped from a single sheet of steel similar to the modern design as shown in figures 1 and 2. Allowing for the accepted standard commercial variations, the steel shall be .0598 inch in thickness. The corners and sides shall be well rounded and flanged for strength. The lower portion of the side frames shall be formed to provide for attachment of the glides. The front edge of the back of the frame between the seat and top back support shall be protected and reinforced by hemming. The side frames shall be prepared with proper holes to permit the attachment of the table arm supports, the seat panel, and the back supports. When specified, the side frame shall, alternatively, be fabricated from 4 pieces of steel angle; two vertical angles shall be of 0.125 minimum thickness, formed and welded into a solid frame meeting all requirements of this specification.

3.2.3.2 *Storage compartment.* — The storage compartment shall be formed from a single sheet of steel 0.0239 inch in thickness, allowing for the commercial variations, and shaped similarly to that shown in the figures. The top front edge shall member neatly with the seat or with the front rail, when used, and the sides shall be secured to the side frames by means of electric resistance welding. Two steel supporting angles or a continuous angle shall be welded to the rear of the compartment to prevent deflections and vibration of the seat. The rear upper edge shall be formed to an angle to house the rear edge of the seat. When the storage compartment forms the front rail, it shall be of 0.0359-inch-thick steel.

3.2.3.3 *Front rail.* — If a one piece storage compartment is used and formed so that the front edge passes under the seat a distance of approximately ½ inch, no front rail is required. The front rail, when used, shall be formed of 0.0747-inch-thick sheet steel, allowing for the commercial variations, and shaped so that it will neatly member with storage compartment. The rail shall be secured by electric resistance welding to the side frames.

3.2.3.4 *Arm supports.*—The arm shall be attached to the chair by means of a front and a rear support similar to those illustrated in the figures. The front supports for the small arm shall be of formed steel 0.1046 inch in thickness. The support for the large arm shall be of 0.1196-inch-thick steel, allowing for commercial variations, and shall be so designed and assembled as to provide a slope of approximately 8 degrees for the arm. The top end of front support for the large arm shall have welded to it a formed V-shaped steel member 0.0749 inch thick, with the top edge flanged not less than ½ inch and the bottom edge flanged and shaped sufficiently to eliminate all sharp edges and corners. The top flanges of the large arm shall have provision for the attachment to the arm with not less than eight 5/8-inch round-head sheet metal screws or equivalent. The top flange of the small arm front support shall be attached to the arm by means of three 5/8-inch round-head sheet metal screws or equivalent. The bottom end of the front supports shall be formed to fit the contour of the chair frame and shall be secured by means of three machine bolts, lock washers and nuts or by welding. The rear supports shall be of formed steel and shall be attached to the right side frame by means of a machine bolt, a lock washer and a nut, or by welding, and to the arm by means of two 5/8-inch round-head sheet-metal screws or equivalent.

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3.2.4 *Wood parts.*

3.2.4.1 *Arm.*—The arm shall be $1\frac{1}{16}$-inch thick 7-ply plywood as described in 3.1.2, or when solid wood is specified, shall be ¾-inch thick by not less than 2-inch wide strips of beech, birch, cherry or maple, and shall be shaped and dimensioned as shown in the figures. The underside of the arm shall have 10 holes drilled in the large table arm and 5 holes in the small arm for attaching the front and back arm supports. The corners shall be well rounded and the edges shall be slightly rounded to minimize damage to the wood and possible injury to the occupant.

3.2.4.2 *Back supports.*—The chair shall have two separate back supports of 5-ply $\frac{7}{16}$-inch plywood, or a single back support 7-ply ¾-inch thick plywood as described in 3.1.2, and shall be formed on a radius of curvature of approximately 18 inches. If two separate back supports are used, the top rail shall be attached to the side frames by means of four solid rivets. The lower back support shall be attached to the side frame in a manner which will permit the rail to pivot and follow the curvature of an occupant's back or be secured by two rivets. If a single back support is used, it shall be fixed or, alternatively, so attached as to pivot and follow the curvature of an occupant's back. The top and bottom edges of the supports shall be rounded and sanded smooth to minimize damage to the wood and to minimize the feeling of sharp edges on the occupant's back.

3.2.4.3 *Seat.*—The seat shall be of plywood not less than 5-ply $\frac{5}{16}$-inch thick before sanding, as described in 3.1.2 and shall be form shaped approximately as shown in the figures. The exposed edges shall be slightly rounded and sanded smooth eliminating all roughness and sharp edges and corners. The seat shall be secured to the side frames by means of four rivets, and shall be neatly and firmly fitted under the flange on the top rear of the storage compartment. The edges of the seat panel shall be inset sufficiently from the frame to protect the edges from possible damage due to bumping during use.

3.3 Weight. — The net weight of the chairs shall not exceed 30 pounds.

3.4 Finish.

3.4.1 *Metal parts.*—All metal parts and surfaces, except exposed rivets, glides, and bolts, shall be thoroughly cleaned and dried and shall have a gray baked-on enamel finish conforming to type B of Federal Specification TT–E–489, or equivalent. The enameled surfaces shall be free from checks, runs, pits, blisters and other defects and shall withstand the adhesion and flexibility test specified in 4.3.3. When specified, color shall conform to number 2630 of Federal Specification TT–C–595.

3.4.2 *Wood parts.* — All exposed wood parts and surfaces shall be finished in medium gloss, clear lacquer or varnish to withstand the wood finish tests specified in 4.3.4 All unexposed surfaces shall receive at least two coats of lacquer sealer or shellac to retard moisture absorption. Exposed finished surfaces, edges and corners shall be free from tackiness, runs, blisters, pits and orange peel and shall be smooth to the eye and touch. The natural color and grain of the wood shall not be clouded by the finishing materials; however, slight toning may be used to accomplish a generally uniform natural color of all wood surfaces on any single item.

3.5 Identification marking.—The manufacturer's nameplate or trade-mark shall be legibly and permanently imprinted or affixed to each chair.

3.6 Workmanship.—The materials used and the methods of construction, design and finishing shall be not less than the requirements of this specification. Chairs shall be free of any defects that impair appearance or which might impair the serviceability of the chair or cause injury or discomfort to the users of the chairs.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for inspection. — Sampling shall be conducted in accordance with Military Standard MIL–STD–105.

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4.2 Inspection.—The chairs shall be subject to inspection by a Government inspector at any time during manufacture or at the point of delivery. Each sample chair shall be inspected to determine compliance with this specification as to workmanship, materials, details of construction, hardware, and finish and tests.

4.3 Tests.—One or more of the sample chairs or at least three truly representative sample specimens shall be subjected to the following tests:

4.3.1 *Plywood shear test.*—Plywood shall be tested in accordance with the shear test requirements of Commercial Standard CS–35.

4.3.2 *Hardwood shear test.*—Hardwood edge joints of tablet arms shall be tested in accordance with the shear test requirements of Federal Specification MMM–A–188.

4.3.3 *Metal enamel adhesion test.*—Specimens of the enameled steel at room temperature shall be bent 180 degrees around a ¼-inch rod without injury to the enamel such as evidence of cracking, chipping, flaking or loss of adhesion.

4.3.4 *Wood finish test.*—The finish of the wood parts shall be tested in the following manner to determine compliance with this specification. There shall be no permanent discoloration, checking or softening of the film surface as a result of the following tests:

4.3.4.1 *Moisture tests.*—Pads of felt saturated with water at approximately 72 degrees Fahrenheit (22 degrees Centigrade) shall be applied to a flat surface of the finish for a period of 48 hours. The pads shall be kept wet during the test period.

4.3.4.2 *Hot water test.*—Boiling water (212 degrees Fahrenheit, 100 degrees Centigrade) shall be allowed to trickle, at the rate of approximately 250 cubic centimeters per minute, over an inclined flat surface for a period of 20 minutes.

4.3.4.3 *Resistance to cleaning compounds test.*—A 5 percent solution of trisodium phosphate shall be applied in contact with the finish of a flat surface for a period of 24 hours.

4.3.4.4 *Wear and abrasion test.*—A block of wood approximately 4 to 6 inches with rounded edges, faced with 1.05-54 sateen and loaded with a total weight of 10 pounds, shall be moved forward and backward across a flat surface. The pad on the block shall be saturated with a paste made with type I polish in accordance with Federal Specification P–P–556. The pad shall be rewet with paste every 50 cycles. The finish shall withstand at least 400 motions in each direction without being cut through to the wood.

4.4 Rejections.—Rejected chairs shall be replaced with other chairs or shall be reworked to correct the defects, after which all specified tests shall be repeated. Rejected chairs shall not be resubmitted for inspection without furnishing full particulars concerning previous rejection and measures taken to overcome the defects.

5. PREPARATION FOR DELIVERY

5.1 For domestic shipment (immediate use).—Unless otherwise specified, subject commodity shall be packed in containers, which (together with packaging, wrapping, and inner packing) are acceptable by common or other carriers for safe transportation to point of destination specified in shipping instructions at the lowest transportation rate for such supplies.

5.2 Domestic shipment (storage).—When specified.

5.2.1 *Packaging.*—The tablet arm and tablet arm support (if detachable) shall be removed from each chair. The attachment hardware shall be placed in a good quality drawstring top, fabric bag and securely tied to the arm support. The arm and support shall be wrapped individually in cushioning material, bundled together with twine ties and securely tied to the back of the back rest. Each chair shall have all surfaces of seat, back rest and legs cushioned with cushioning pads made up with the following materials or combinations thereof:

Macerated paper	Creped cellulose wadding
Shredded paper	Excelsior (sealed ends)
Wood wool	Cotton wadding
Wood fiber felt	Molded pulpboard

Laminated creped cellulose wadding backed with corrugated fiberboard.

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Cushioning pads shall provide built-up cushioning of not less than 3/8 inch in thickness. All cushioning shall be secured by spirally tying with not less than two-ply jute twine conforming to Federal Specification T-T-911, type B. Two cushioned chairs shall be nested as follows: One chair shall be inverted on a second chair with the seats of the two chairs together, the backs on opposite sides and the legs at opposite ends. The front legs and back posts which adjoin shall be tied together with twine having a breaking strength of not less than 150 pounds.

5.2.2 *Packing.*—The two nested chairs shall be packed in a snug-fitting box conforming to Federal Specification LLL-B-631, LLL-B-636, NN-B-591, PPP-B-601, NN-B-621, or PPP-B-585.

5.3 For export shipment (commercial).—When specified, subject commodity shall be packed in containers which, together with packaging, wrapping and inner packing, are acceptable by oceangoing vessels for safe transportation to the points of destination specified in the shipping instructions at the lowest transportation rate for such supplies.

5.4 For export shipment (military).—When specified.

5.4.1 *Packaging.*—Chairs shall be packaged as specified in 5.2.1.

5.4.2 *Packing.*—The two nested chairs shall be packed in a snug-fitting box conforming to Military Specifications JAN-P-108, JAN-P-103, JAN-P-106, MIL-B-107, JAN-P-138, or Federal Specification PPP-B-601.

5.5 Marking.

5.5.1 *Non-military.*—Unless otherwise specified, shipping containers shall be marked with the name of the commodity, the class and color, the Federal stock number, the quantity contained therein as defined by the contract or order under which the shipment is made, the name of the contractor, the number of the contract or order, and the full name and address of the consignee.

5.5.2 *Military.*—Marking shall be in accordance with Military Standard MIL-STD-129. The item name shall be "Chair, Student, Metal and Wood (Class ____, Color ____)."

6. NOTES

6.1 Ordering data.—Purchasers should specify the class of chair required and should exercise any desired options offered herein (see 1.2.1, 1.2.2, 3.2.3.1, 3.4.1, section 5 and 6.3).

6.2 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal government requirements.

6.3 Preproduction sample. — When specified and before production is commenced, a complete unit shall be made available to the contracting officer for his approval, either at the contractor's plant or at such other place as designated. When the preproduction sample is approved, it shall be retained by the inspector as a control guide for subsequent production on the contract. The approved sample may be included as a part of the last shipment.

6.4 The proper transportation description and carload and truckload weights are as follows:

Chairs, school, metallic or wooden, with tablet arms or with tablet arm attachments, set up.

In cars 40'9" or less in length, inside measurements:

Carload weights 10,000; 12,000; 15,000; 18,000; 22,000; 30,000 pounds.

In cars exceeding 40'9" in length, inside measurements:

Carload weights 16,000; 18,000; 21,000; 24,000; 30,000; 40,000 pounds.

Chairs, school, metallic or wooden, with tablet arms or tablet-arm attachments, set up.

Truckload weight 16,000 pounds.

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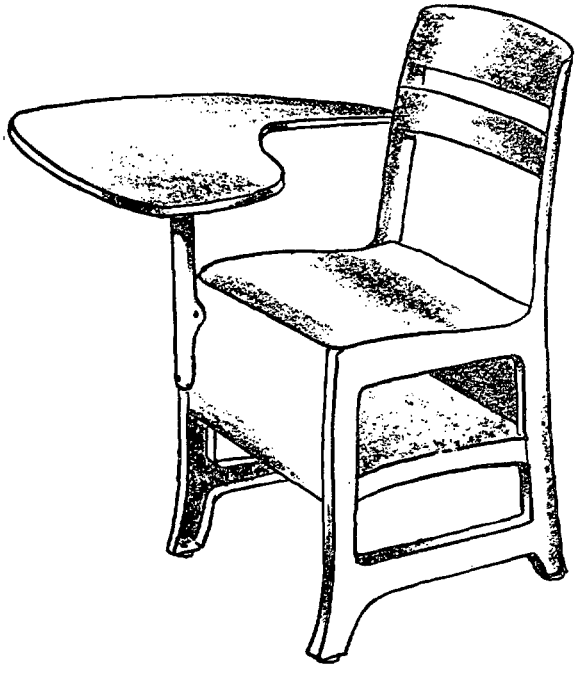
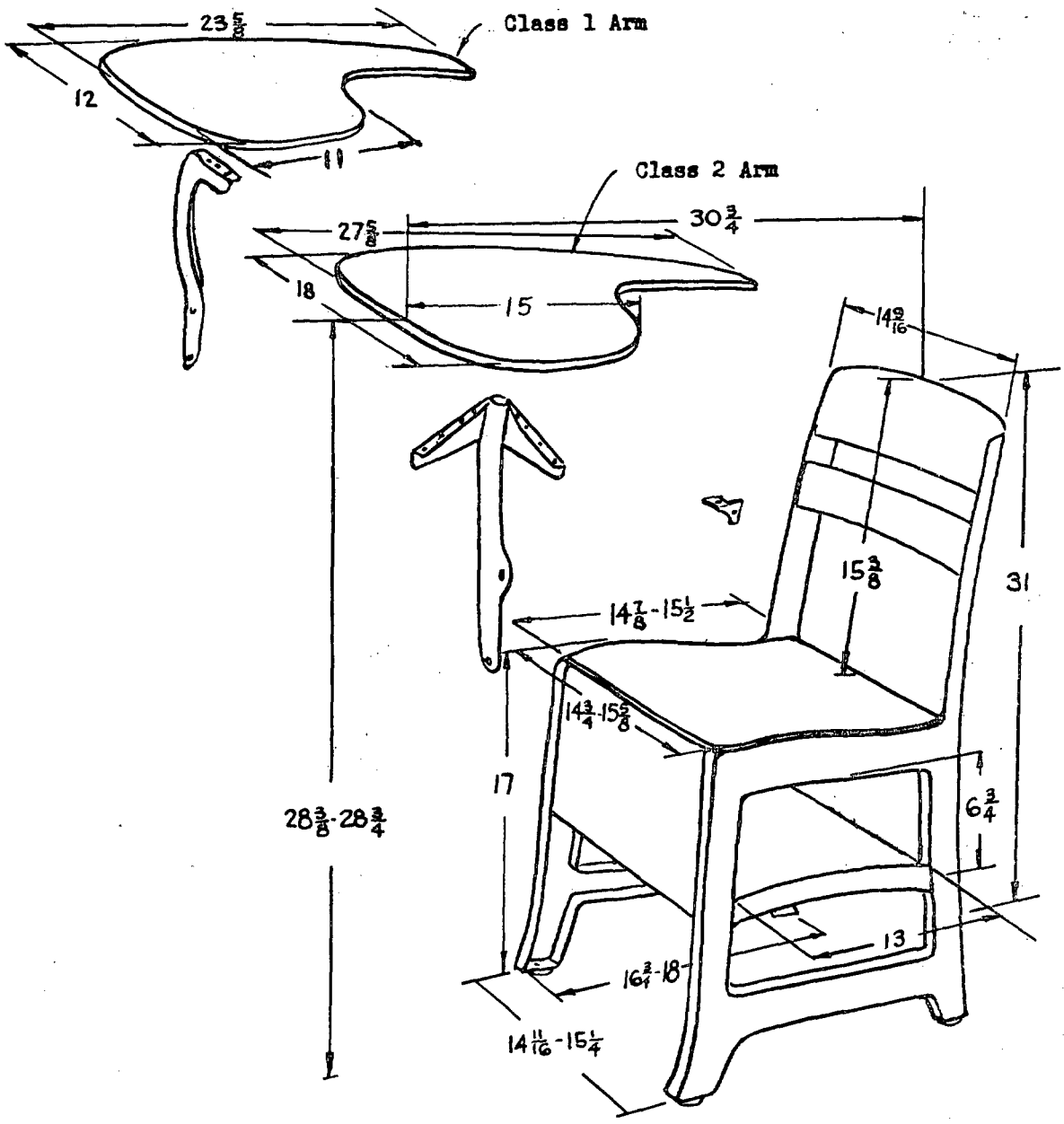


FIGURE 1.—Chair, student, metal and wood.

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Dimensions In Inches

CHAIR, METAL AND WOOD, TABLE ARM

Figure 2

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